

I Have a Stream: Making Self-supervised Learning Work on Continuous Video

Ivan Martinović · Lukas Knobel · Yuki M. Asano

Motivation

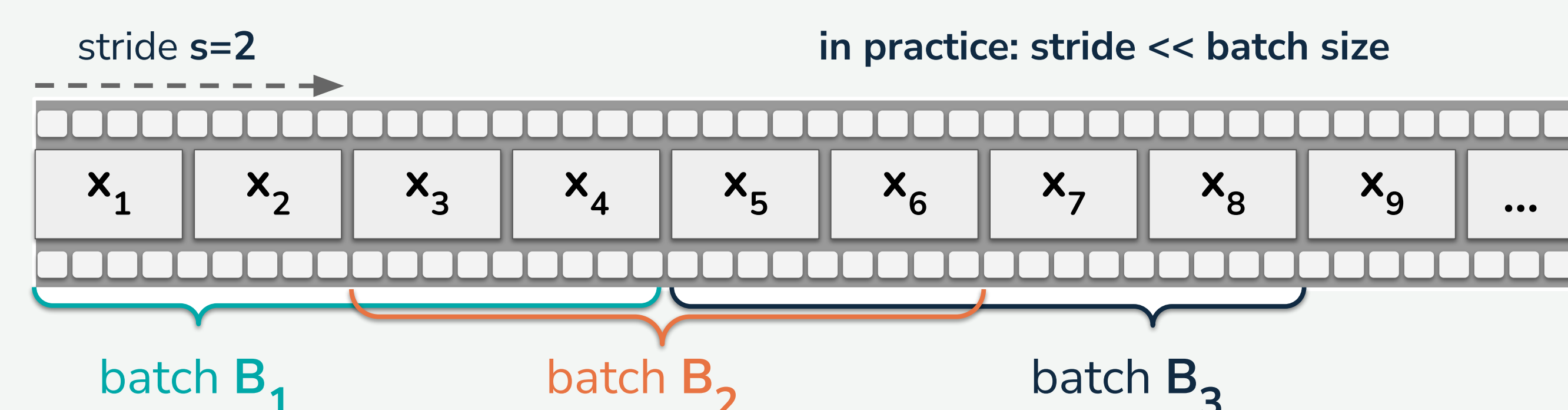
Natural visual experience is sequential, whereas standard SSL relies on globally shuffled I.I.D. data.



Learning from a Continuous Video Stream

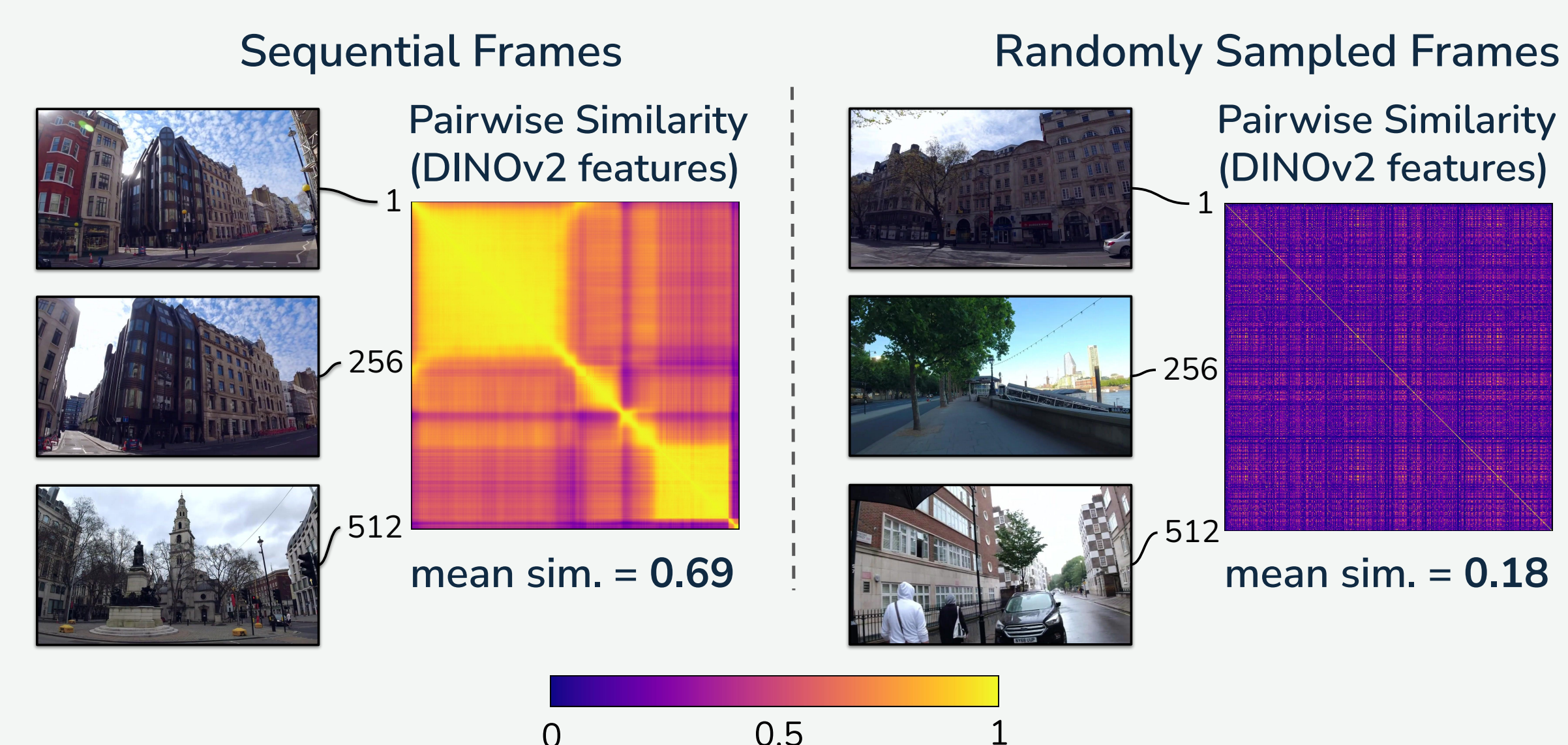
Streaming Data & Setup

We construct **WalkingTours++**, a video stream of up to **95 hours**, by concatenating 58 walking-tour videos.



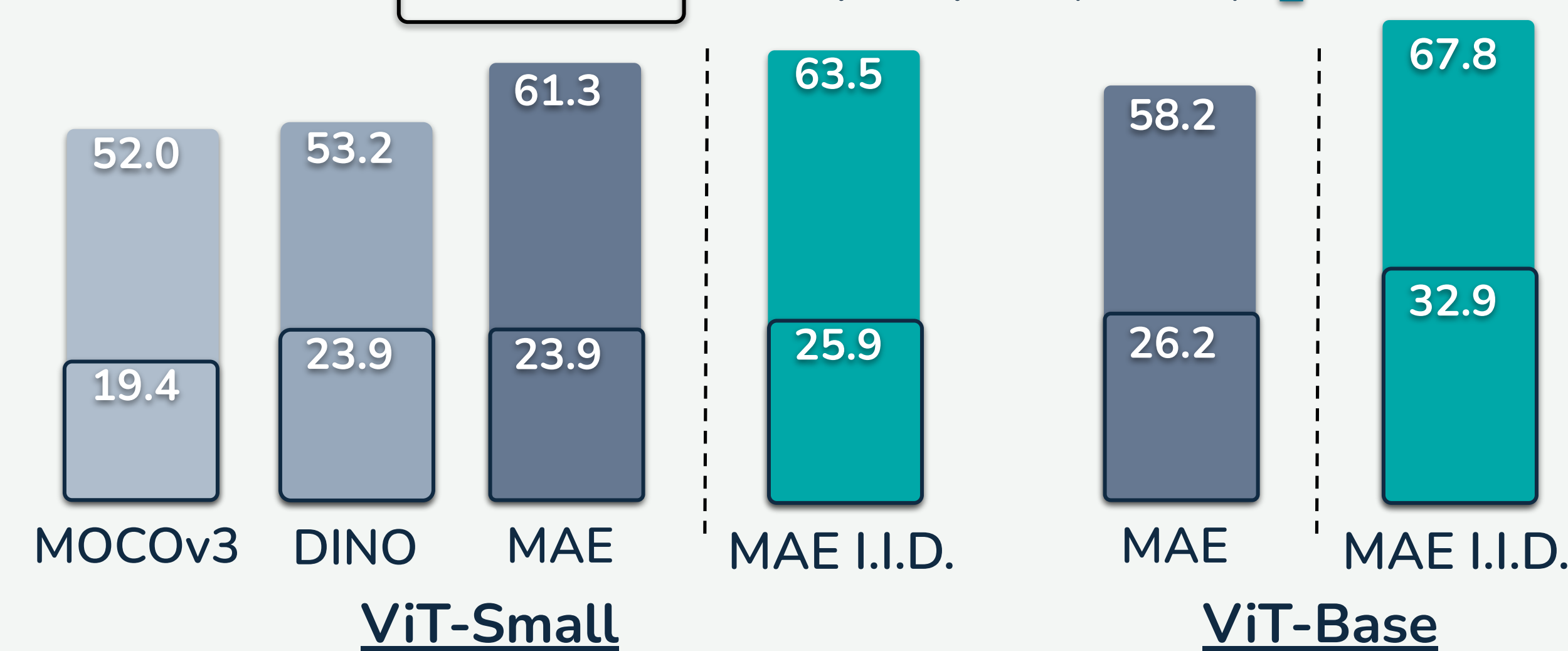
Can SSL work **well** on continuous video streams, starting from **random init**, **without global shuffling**, **repeated passes**, and **large replay buffers**?

Streams exhibit high similarity within each batch.



MAE is the strongest baseline under streaming setup, but still trails **I.I.D.-trained MAE** on the same data.

ADE20K & Cityscapes (mIoU) ↑



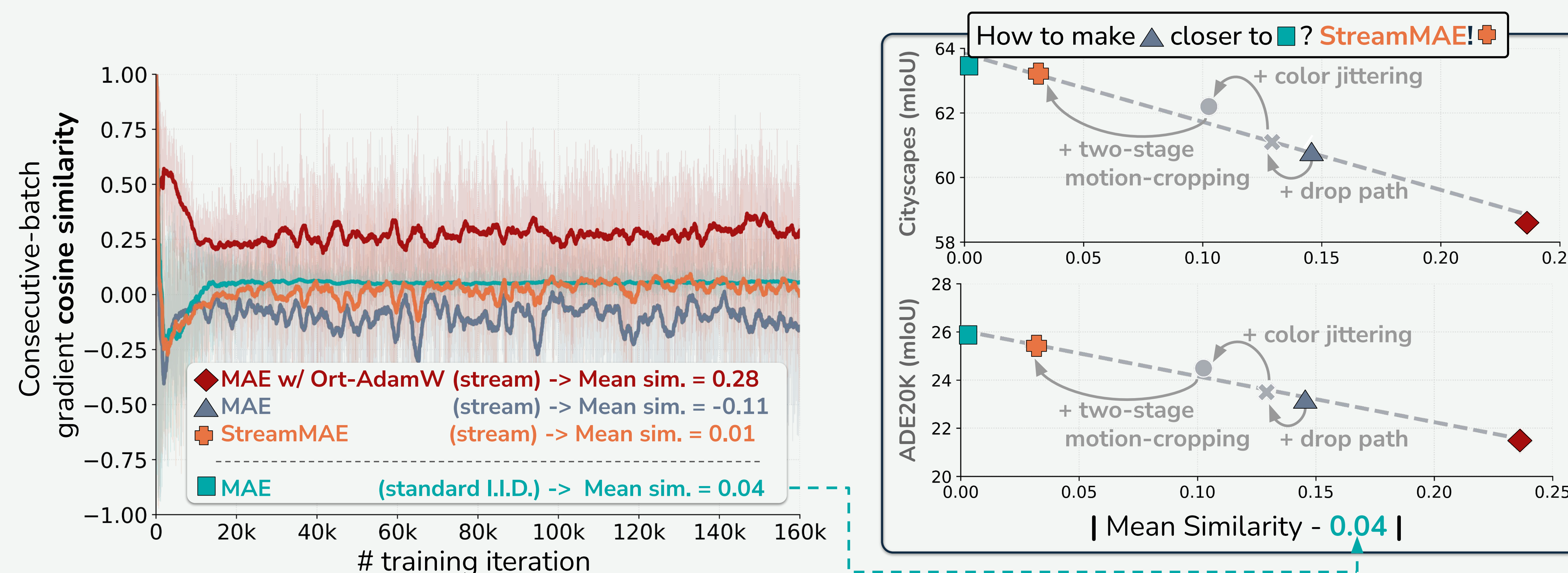
Why Does Streaming MAE Lag Behind I.I.D. MAE?

High inter-batch similarity (overlap between consecutive batches) does not explain the gap.
High intra-batch similarity (near-duplicate frames within each batch) is the main challenge.

ImageNet-1k				
training regime	μ_{inter}	μ_{intra}	City mIoU	ADE mIoU
I.I.D.	0.004	0.325	64.0	26.9
pre-shuffled stream	0.004	0.989	63.6	27.2

WT++12h				
training regime	μ_{inter}	μ_{intra}	City mIoU	ADE mIoU
I.I.D.	0.171	0.738	67.8	32.9
pre-shuffled stream	0.171	0.999	68.8	32.3
chronological stream	0.665	1.000	58.2	26.2

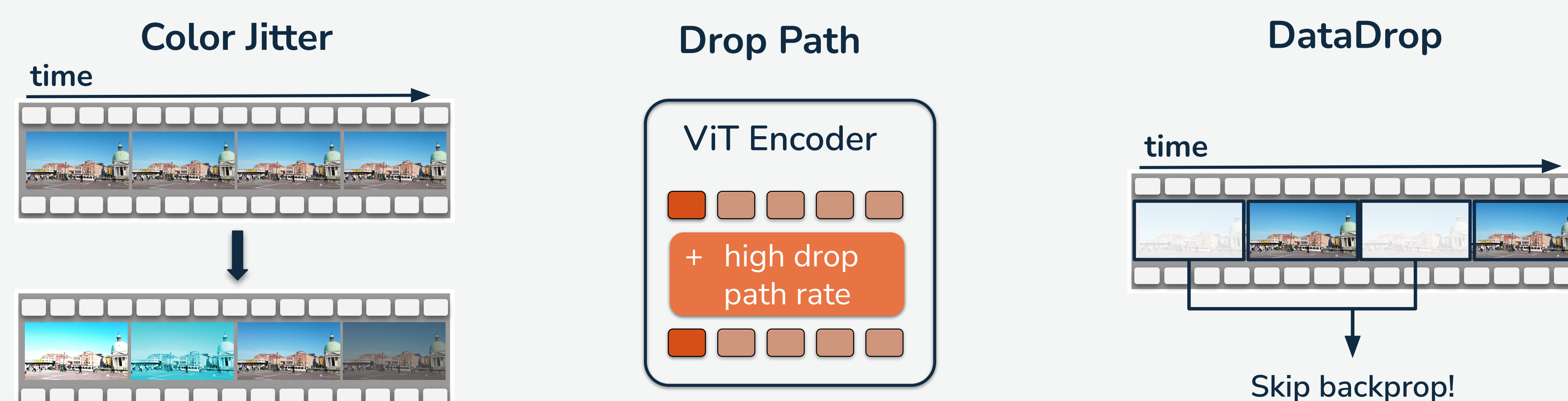
High intra-batch similarity shifts update dynamics away from the I.I.D. regime.



StreamMAE

StreamMAE retains the standard MAE objective while introducing **stream-tailored regularization** and **motion-biased crop selection**.

Stream-tailored regularization



Motion-biased crop selection

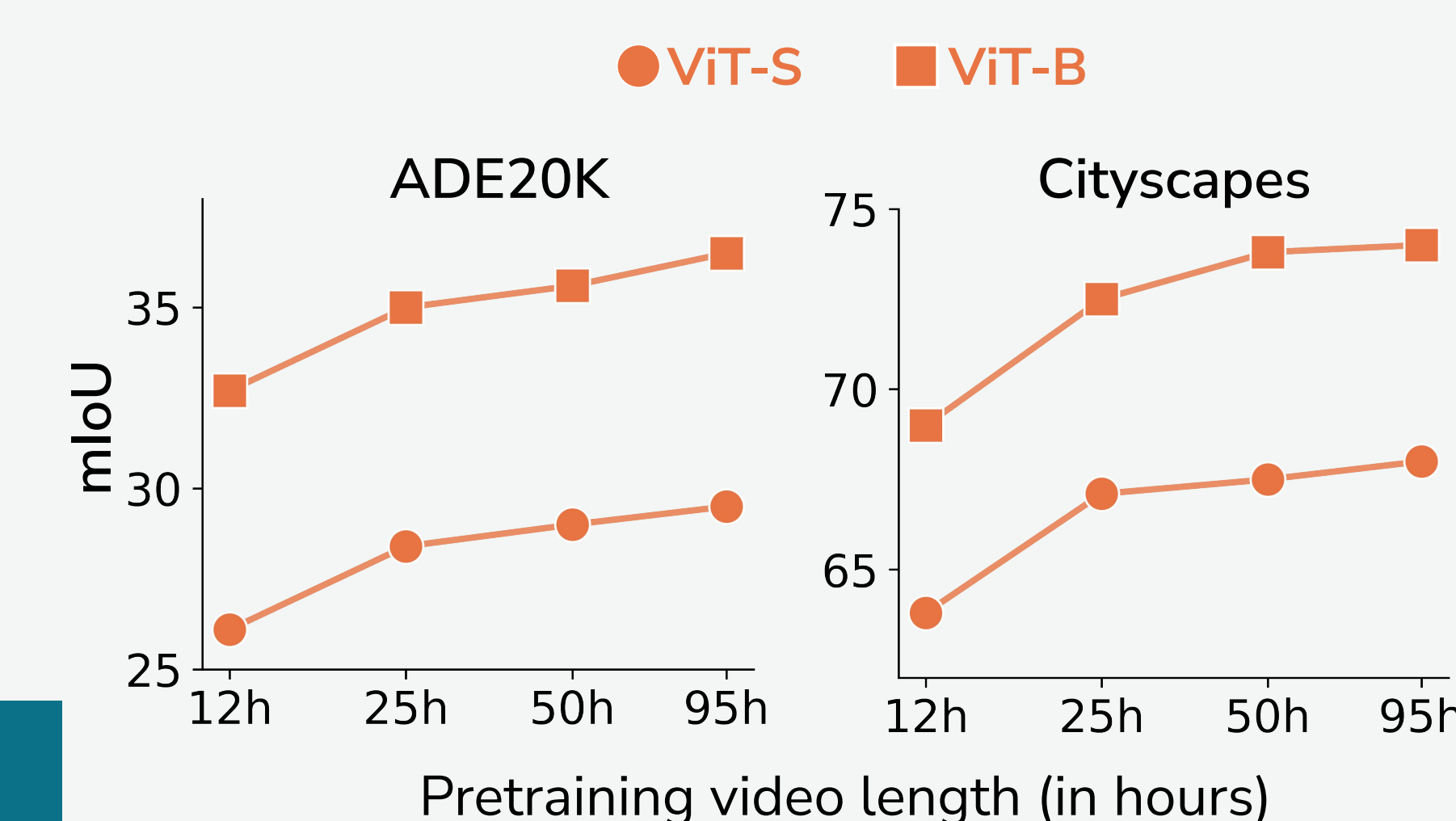


Evaluating StreamMAE

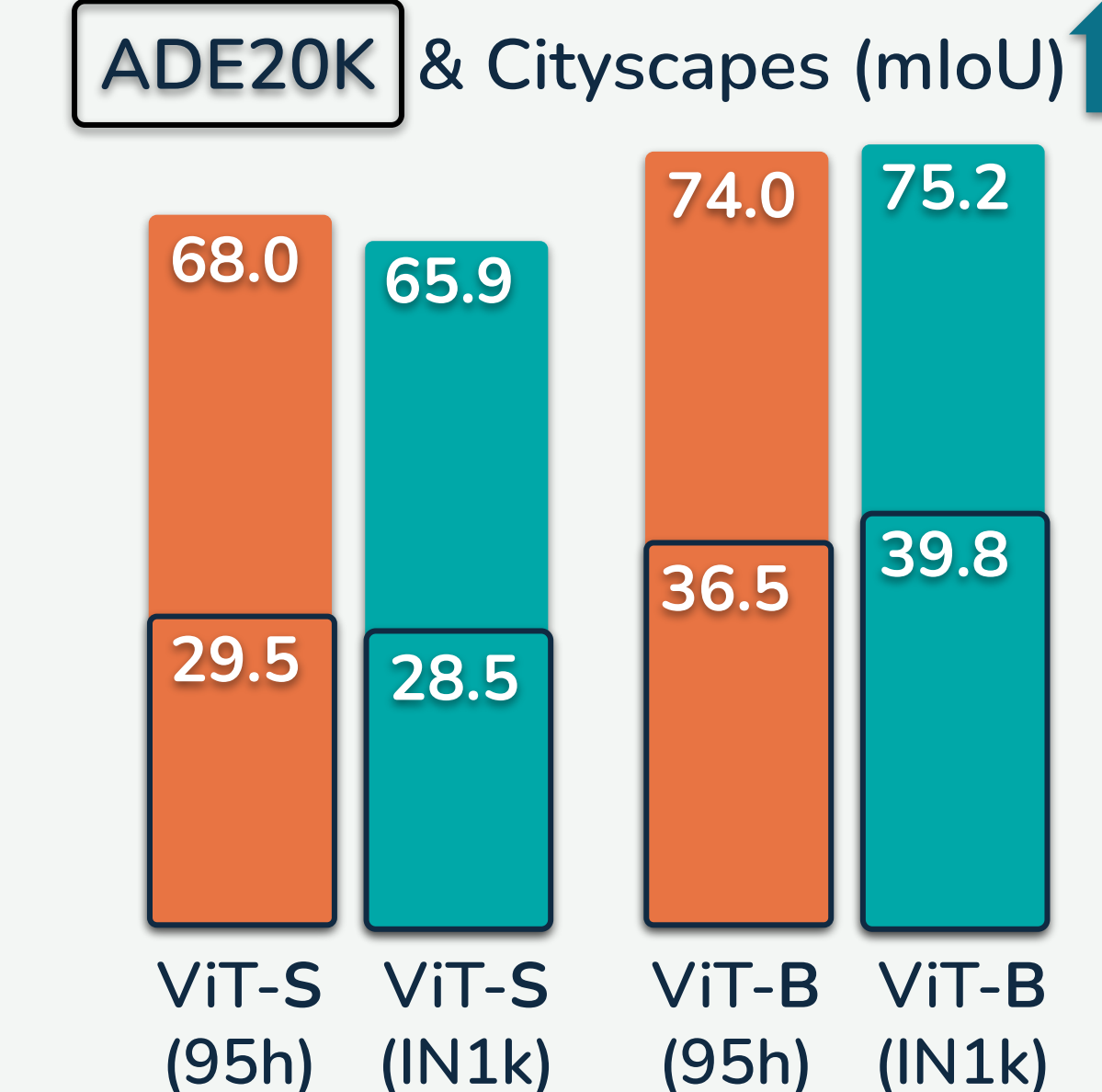
StreamMAE outperforms all streaming baselines using the same pretraining data and **matches I.I.D.-trained MAE**.

Pretraining Method	Pretraining Data	IN1K acc@1	City mIoU	ADE mIoU	NYU2 RMSE	KITTI RMSE
Random Init	–	71.9	47.7	16.7	0.877	6.108
MOCO v3	WT++12h	68.7	52.0	19.4	0.760	4.882
DINO	WT++12h	72.7	53.2	23.9	0.734	4.454
MAE	WT++12h	77.1	61.3	23.9	0.742	4.211
Orthogonal-MAE	WT++12h	75.9	58.3	22.5	0.749	4.209
MemoryStoryboard	WT++12h	71.0	49.2	21.6	0.792	4.647
StreamMAE (ours)	WT++12h	77.5	63.8	26.1	0.694	3.976
MAE - standard I.I.D.	WT++12h	77.0	63.5	25.9	0.701	4.070

StreamMAE improves with longer streams and a larger model.

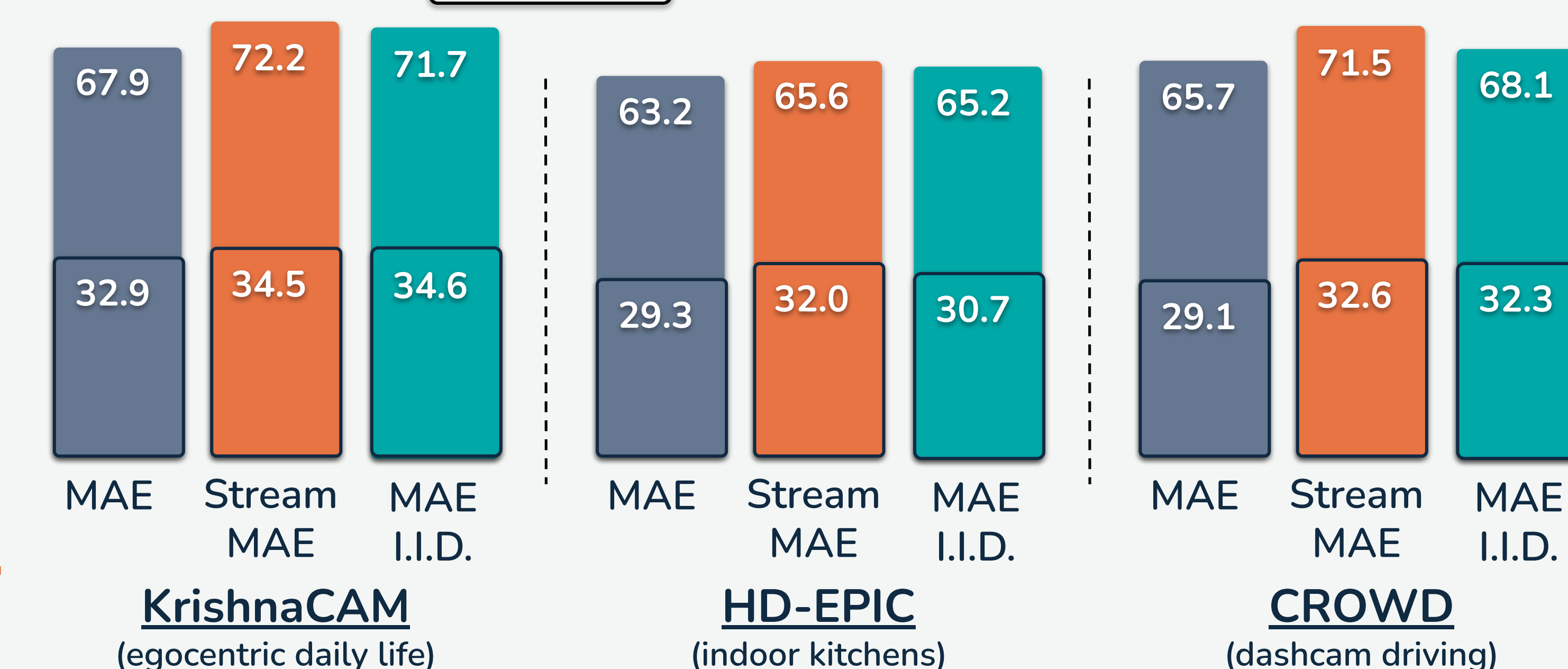


StreamMAE approaches MAE trained on IN-1k



StreamMAE improves over MAE across video domains.

ADE20K & Cityscapes (mIoU) ↑



Conclusion

We identify high intra-batch similarity as the main challenge in streaming SSL. **StreamMAE** addresses this redundancy and matches I.I.D.-trained MAE on the same data.

Project Page

